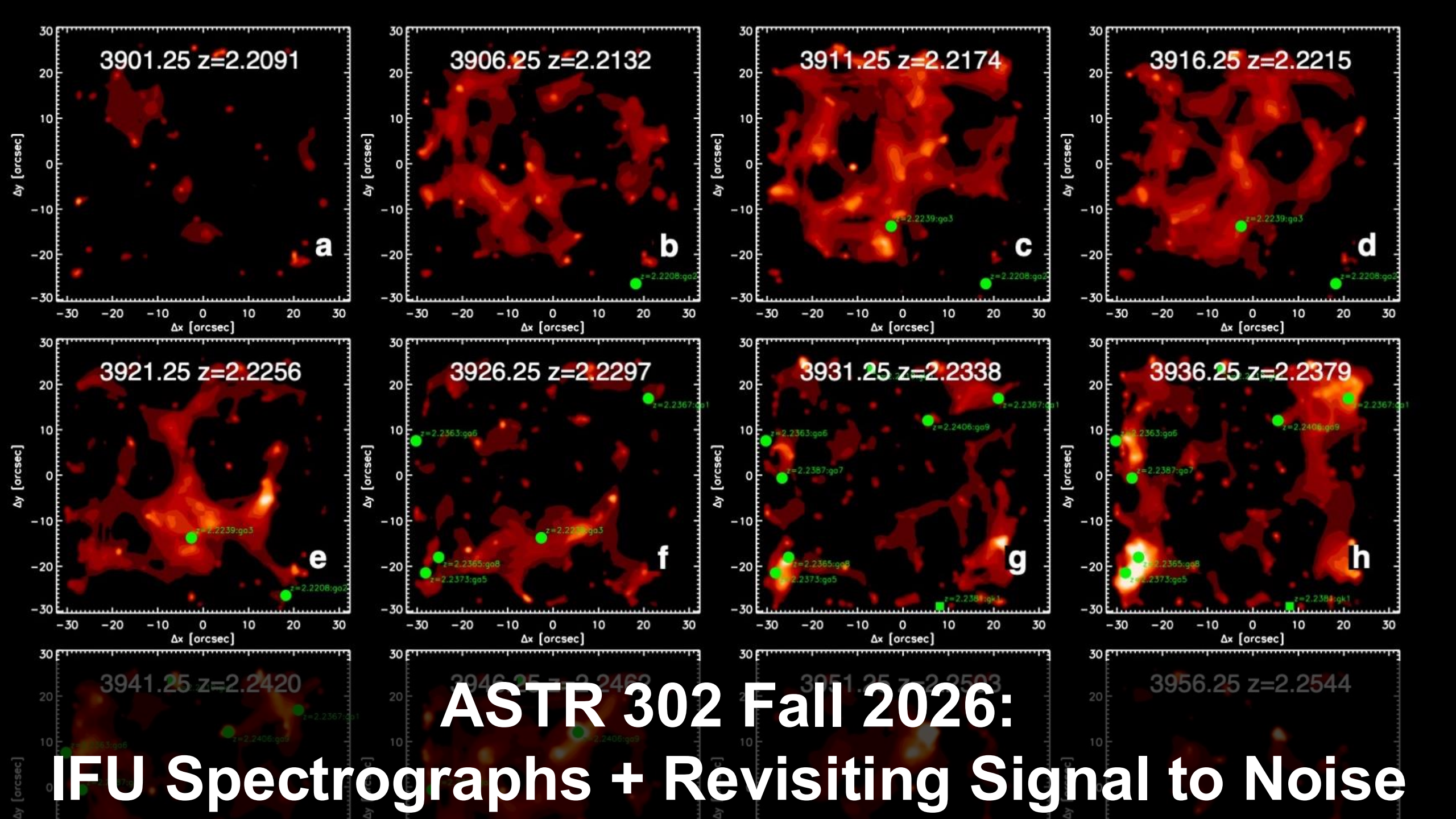


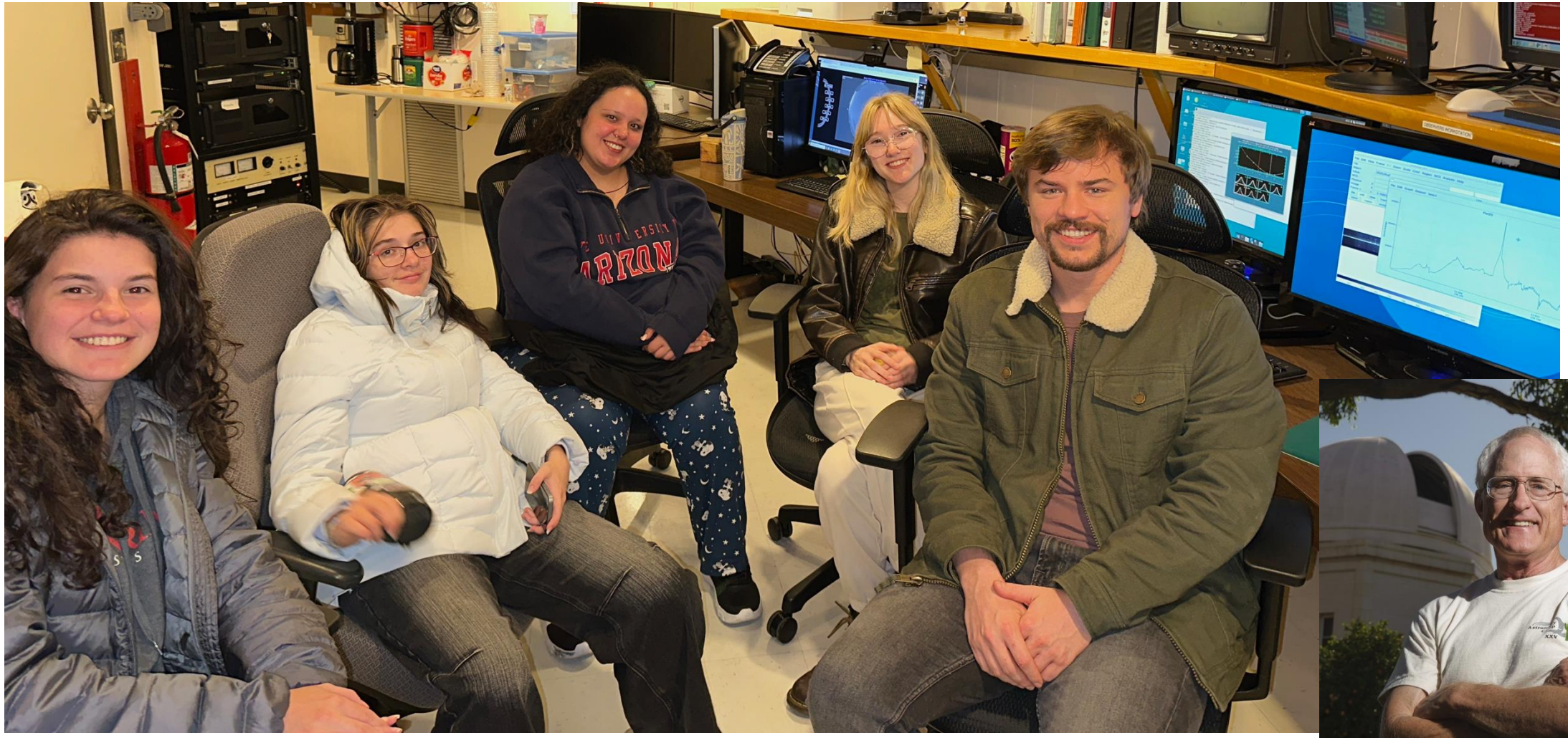
ASTR 302 Fall 2026:
IFU Spectrographs + Revisiting Signal to Noise

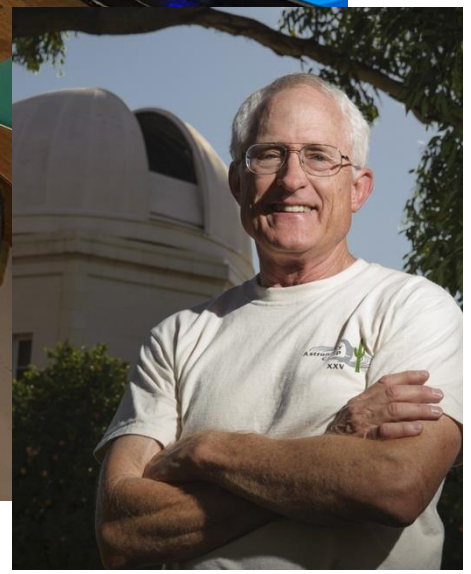
Agenda – October 19, 2026

- **Observing RECAP**
- **Dr Miles will be in class next week**
- **Integral Field Spectrographs (Imaging + Spectroscopy)**
- **Estimating Signal-to-Noise + Putting it All Together**
  - From “known” sources of error (CCD Equation)
  - From your own data

Recommended Reading: Birney Chapter 9, 10

OBSERVING RECAP:
❗🚨❗SOMEBODY FINALLY GOT DATA LIVE❗🚨❗



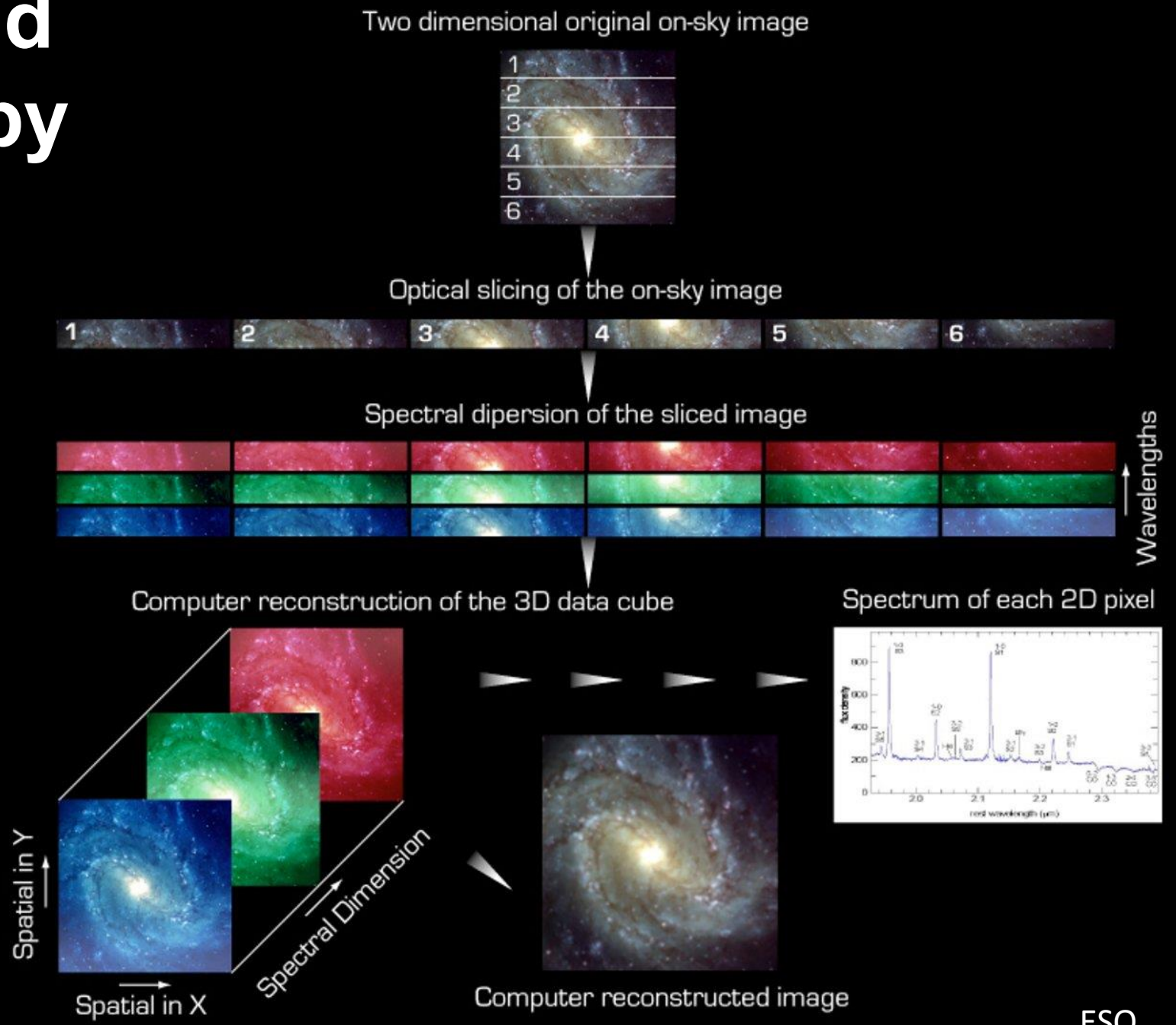


Bok Wine Supernova….BOOM! + Collin Christy
Photo by Don McCarthy

Integral Field Spectroscopy

Integral Field Spectrographs (IFS) or Units (IFU) combine the ability of imaging and spectroscopy.

These imaging + spectroscopic instruments allow you to efficiently get spatial and spectral imaging at the same time.



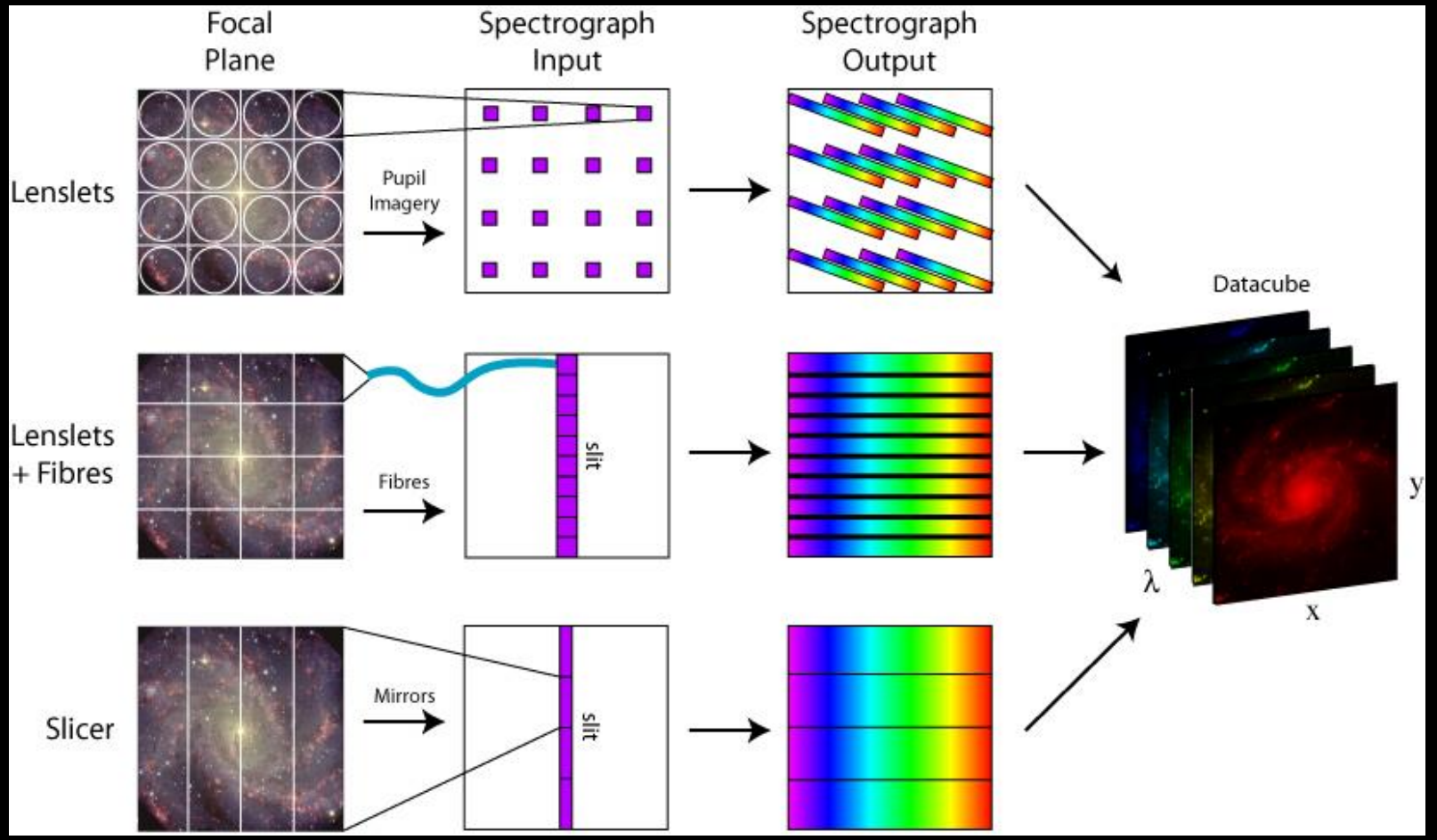
ESO

Integral Field Spectroscopy: Creating Pseudo Slits

Lenslet arrays use a series of lenses (often tiny) to break up a field of view into subapertures.

Fibers can be placed behind lenslets to efficiently pack spectra.

Fibers can be used with lenslet arrays.

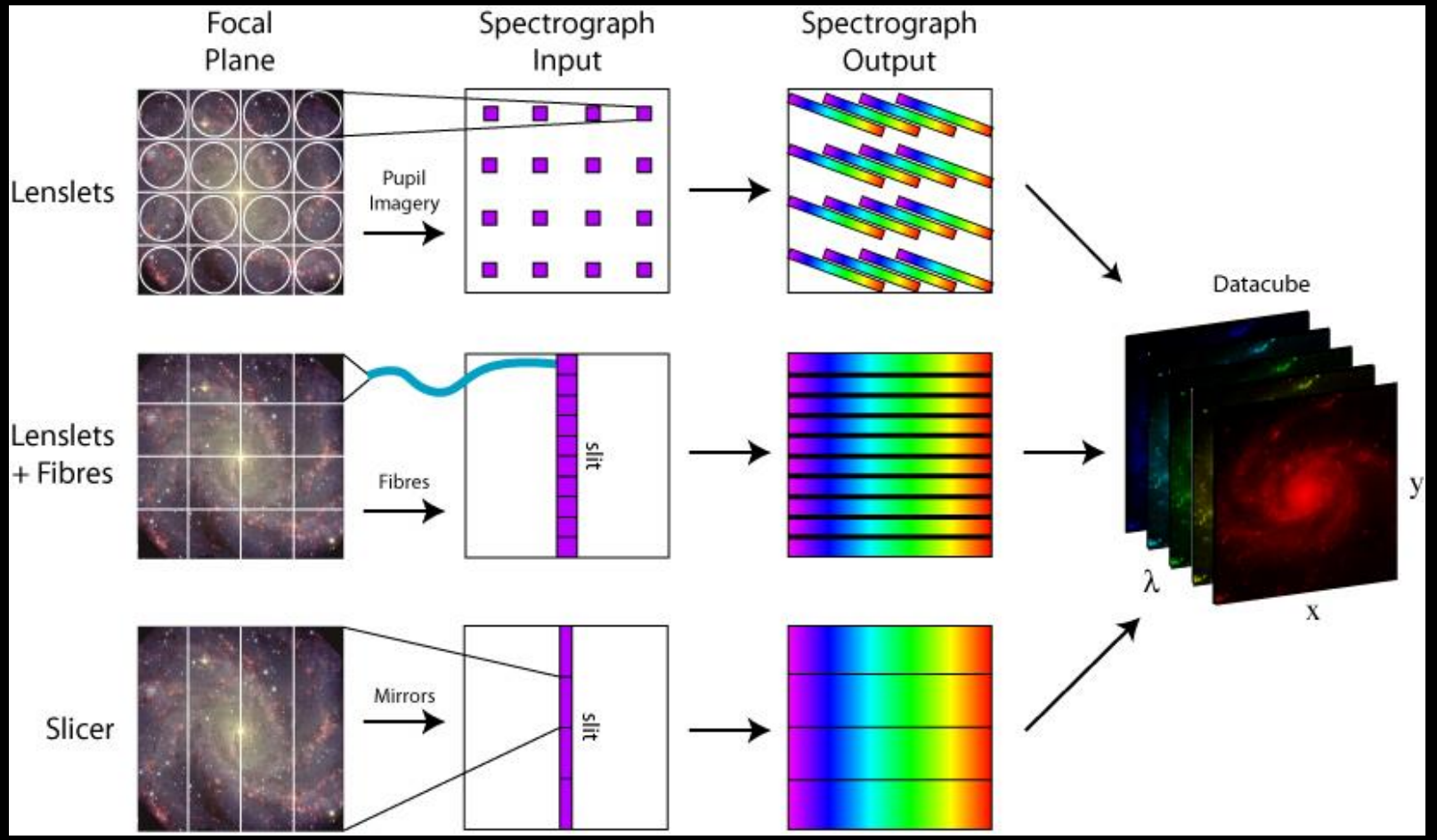


ESO

Integral Field Spectroscopy: Creating Pseudo Slits

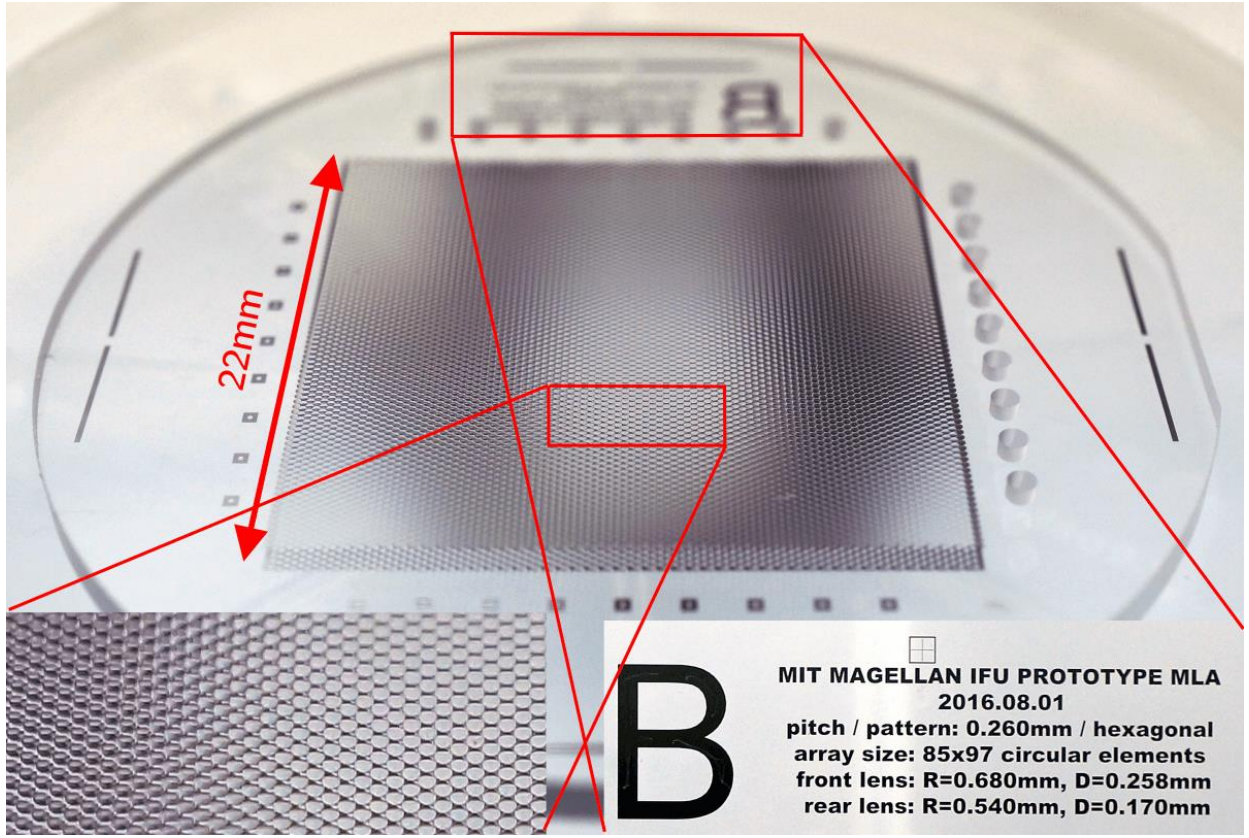
There are two primary ways a field of view can be broken up into smaller slits or pseudo slits.

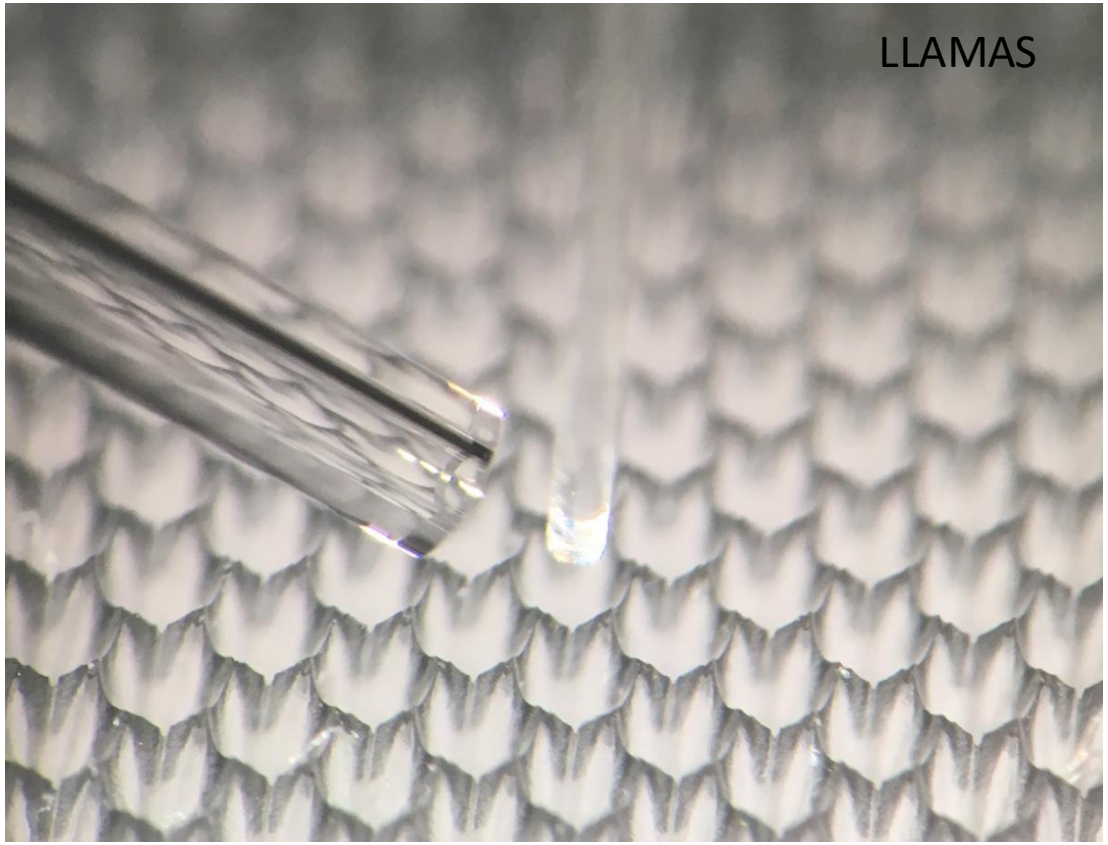
Image Slicers use mirrors to break up the field of view in “slices” or slits to then be dispersed.



ESO

Integral Field Spectroscopy: Lenslet + Fibers

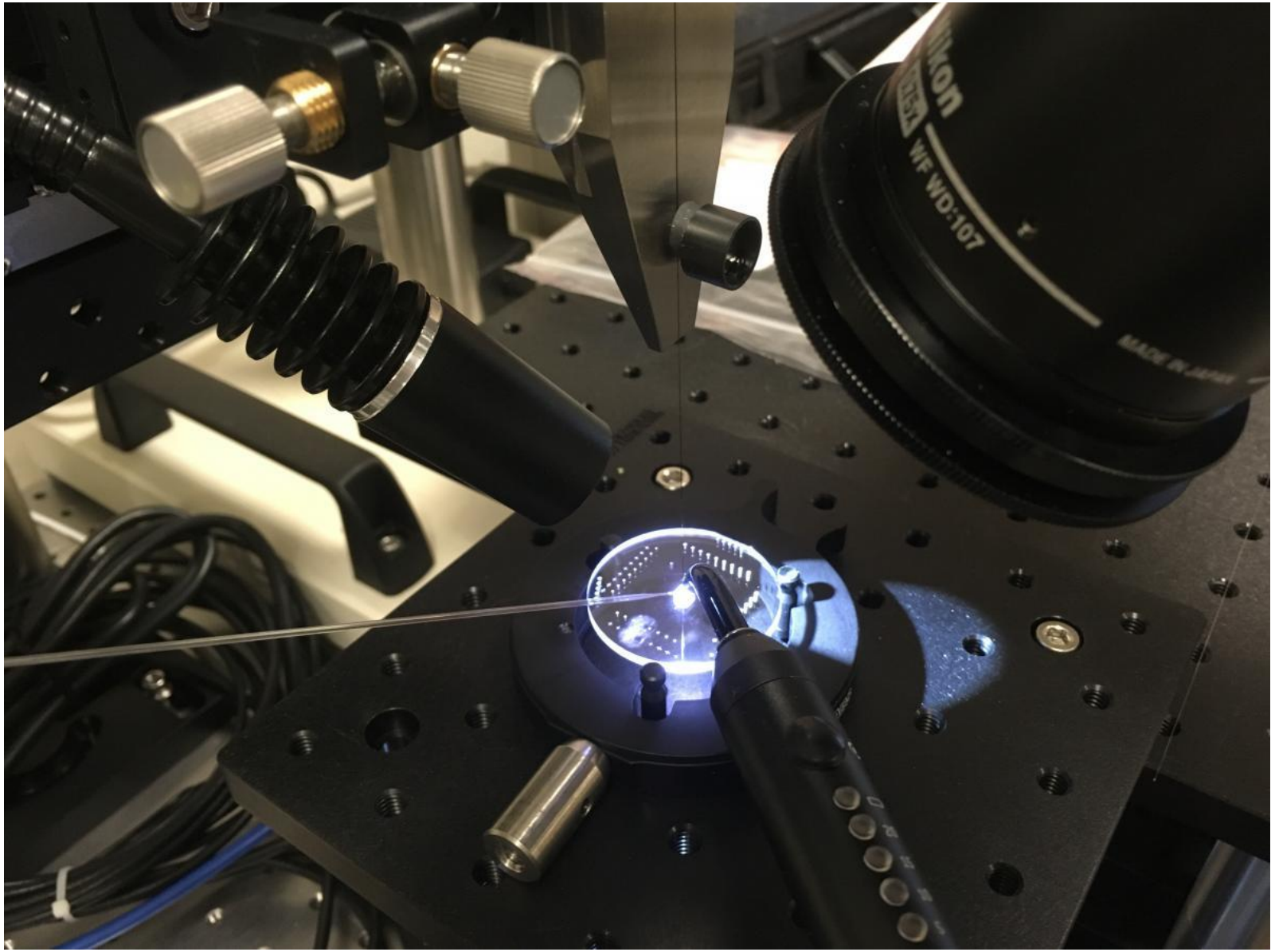




- Companies can etch a lenslet pattern into a block of material for astronomers. This process is not always precise and is prone to errors.
- Placing fibers behind lenslets can be tedious. Fibers are not often used cryogenically (YET!)

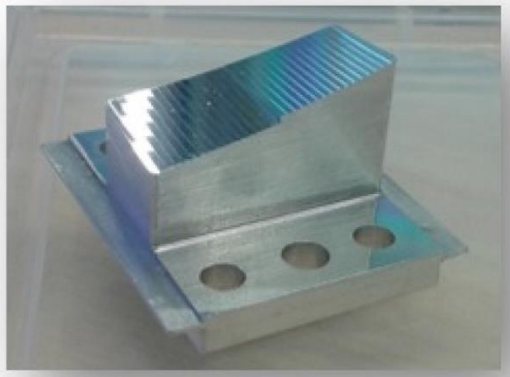
https://ait.mit.edu/instruments/llamas/

Integral Field Spectroscopy: Lenslet + Fibers



https://ait.mit.edu/instruments/llamas/

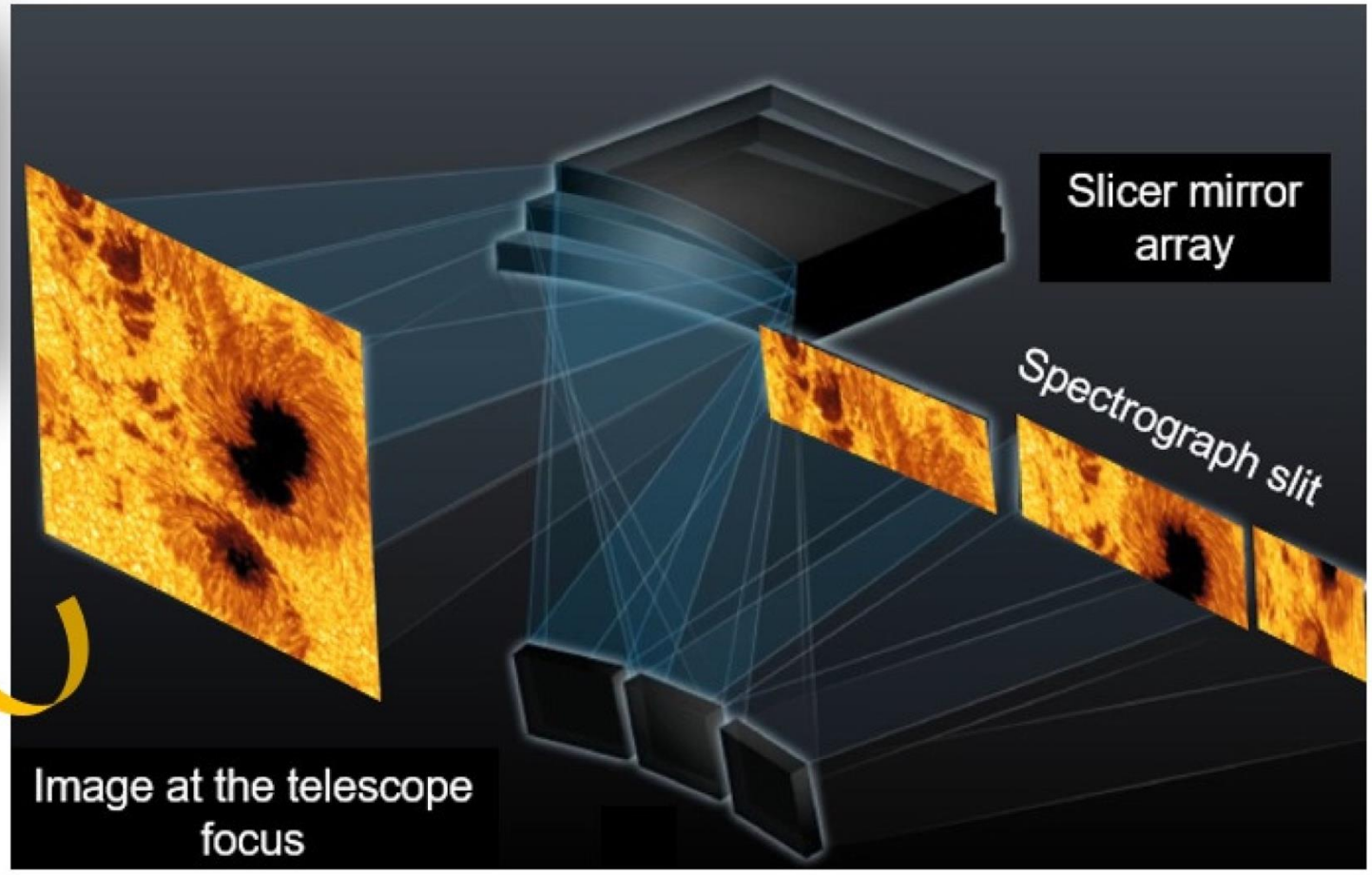
Integral Field Spectroscopy: Image Slicers



Slicer mirror array

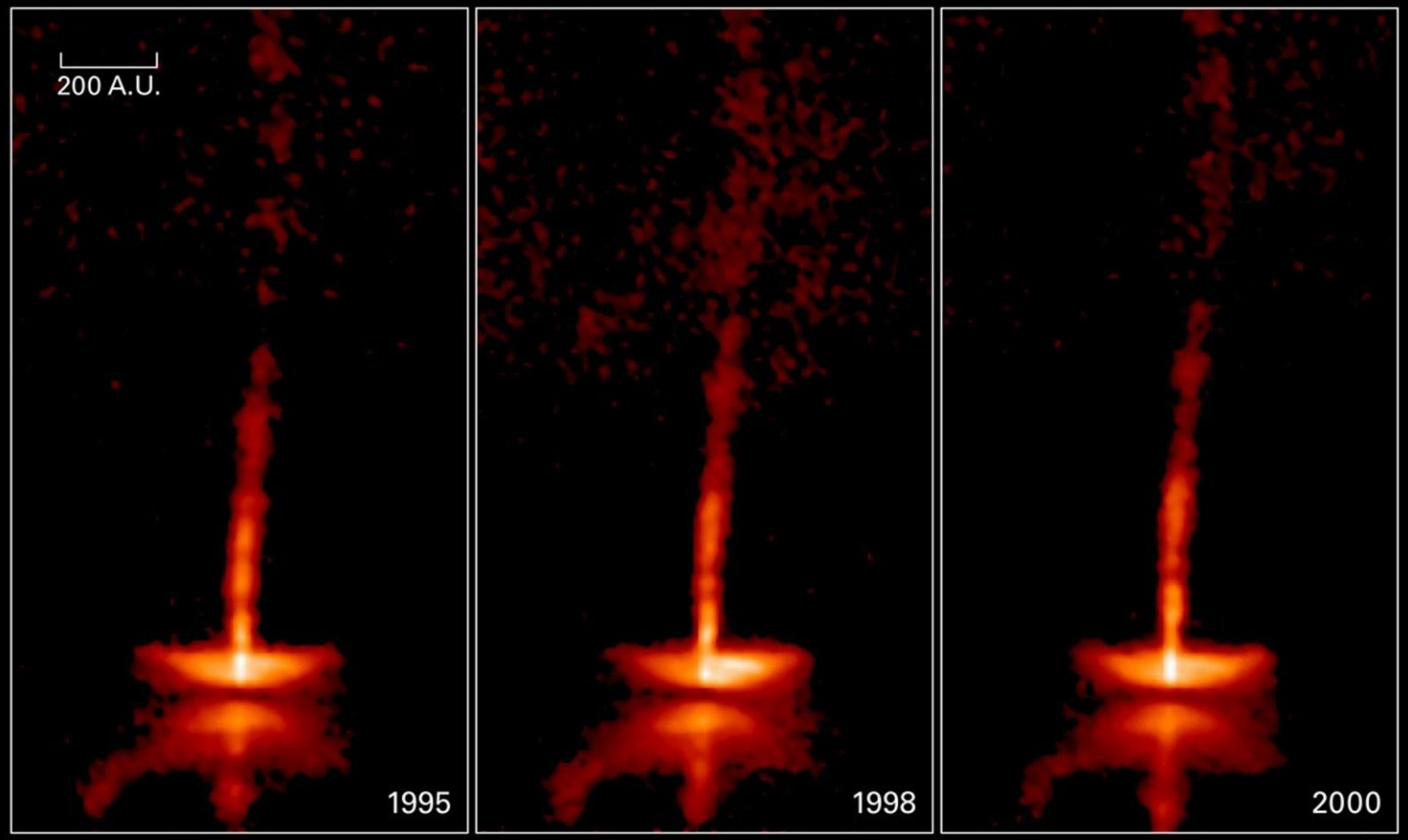


Sliced FOV



- Reflective
- Expensive
- Works great at room temperature and cryogenically.

The Power of Integral Field Spectroscopy



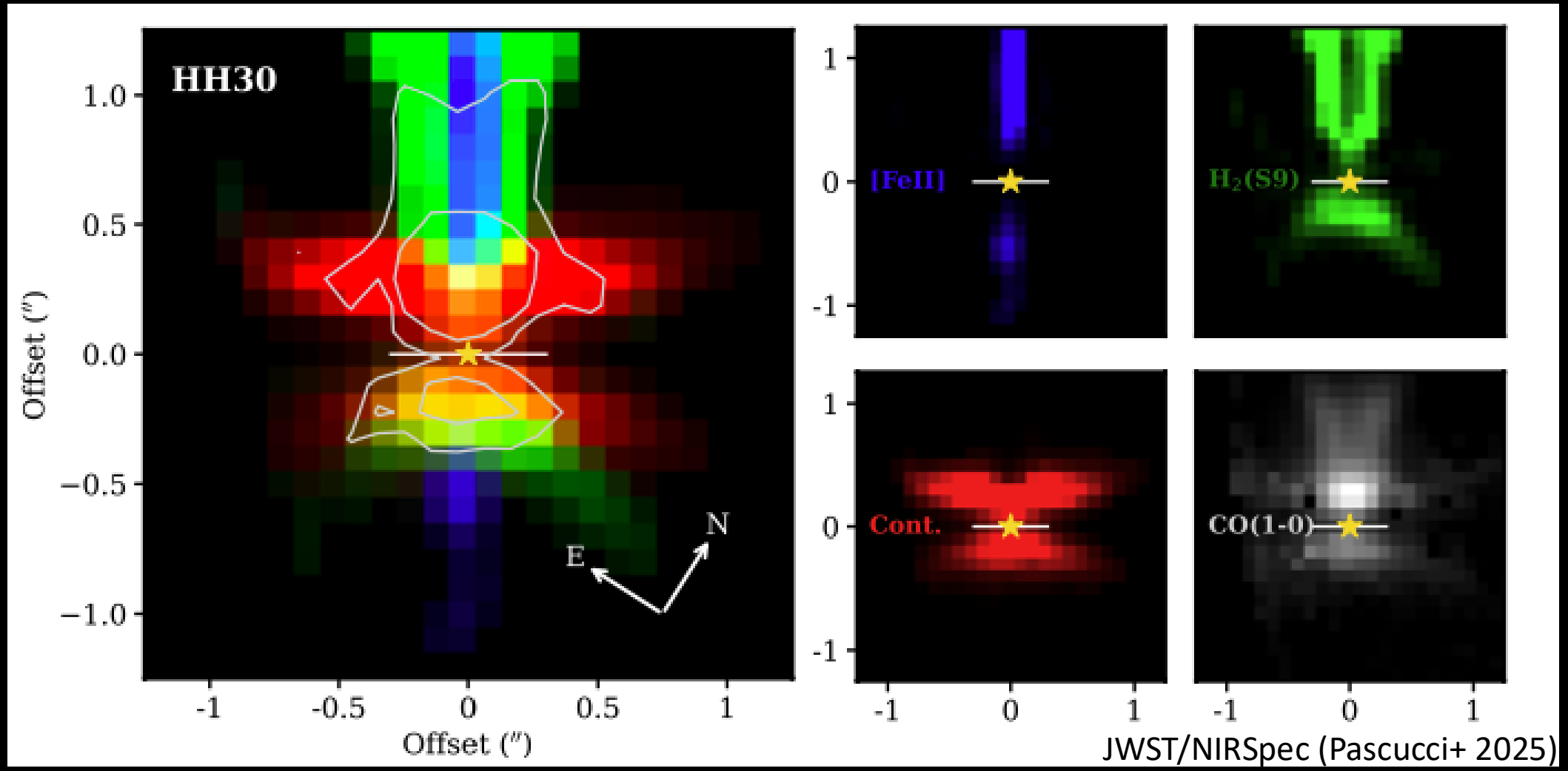
The Dynamic HH 30 Disk and Jet
Hubble Space Telescope • WFPC2

NASA and A. Watson (Instituto de Astronomía, UNAM, Mexico) • STScI-PRC00-32b

Herbig-Haro (HH) Object 30

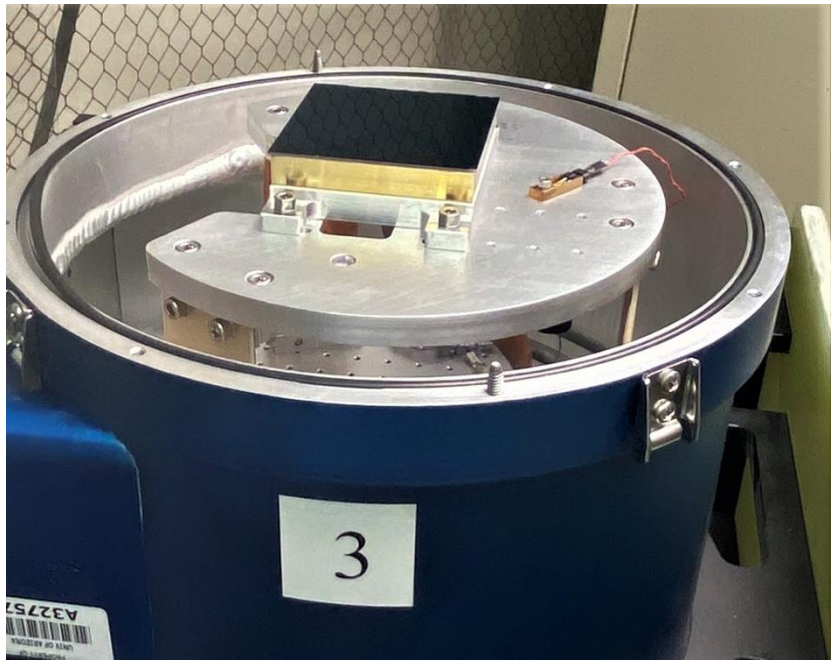
A young star with a protoplanetary disk. This system is in the early stages of planetary formation.

The Power of Integral Field Spectroscopy



IFS instruments allow us to understand the spatial extent of different molecules.
If the resolution of your IFS is high enough, you can estimate the speed of flows.

Noise Sources in CCDs/Detectors



90 prime CCD exposed in dewar/cryostat.

CCDs introduce both **random** and **systematic** noise into our observations. Therefore, CCDs impact both the precision and accuracy of our measurements.

To mitigate this, we must characterize and understand several important properties.

- **Bias** – Electronic Noise Offset
- **Dark Current** – Thermal Noise Offset
- **Flat Field** – Per pixel response uniformity
- **Read/Poisson Noise** – Counting precision

Signal to Noise (SNR)

$$SNR = \frac{gS_{star}}{\sqrt{gS_{star} + ngS_{sky} + ngS_{dark} + nS_{read}^2}}$$

S_{star} : Signal from the star (ADU)

S_{sky} : Signal from the sky (ADU)

S_{dark}: Signal from the dark (ADU)

S_{read}: Read noise (e-)

g: gain (e-/ADU)

n : Number of pixels

This equation does not include every possible error term, but the main ones for CCDs

Question: While observing at the Kuiper-61" with the Mont4K camera, you measure two stars: a science star (S) and a reference star (R). Using a 6 pixel radius aperture, you measure 10,000 ADU for the target star and 12,000 ADU for the reference star in a 90 sec exposure. Calculate the SNR for each star using the SNR equation.

Gain: – 3.1 e-/ADU
Dark Current: 16.6 e-/pixel/hour
Sky Background: 0.1 e-/pixel/s
Read Noise: 5.0 e-

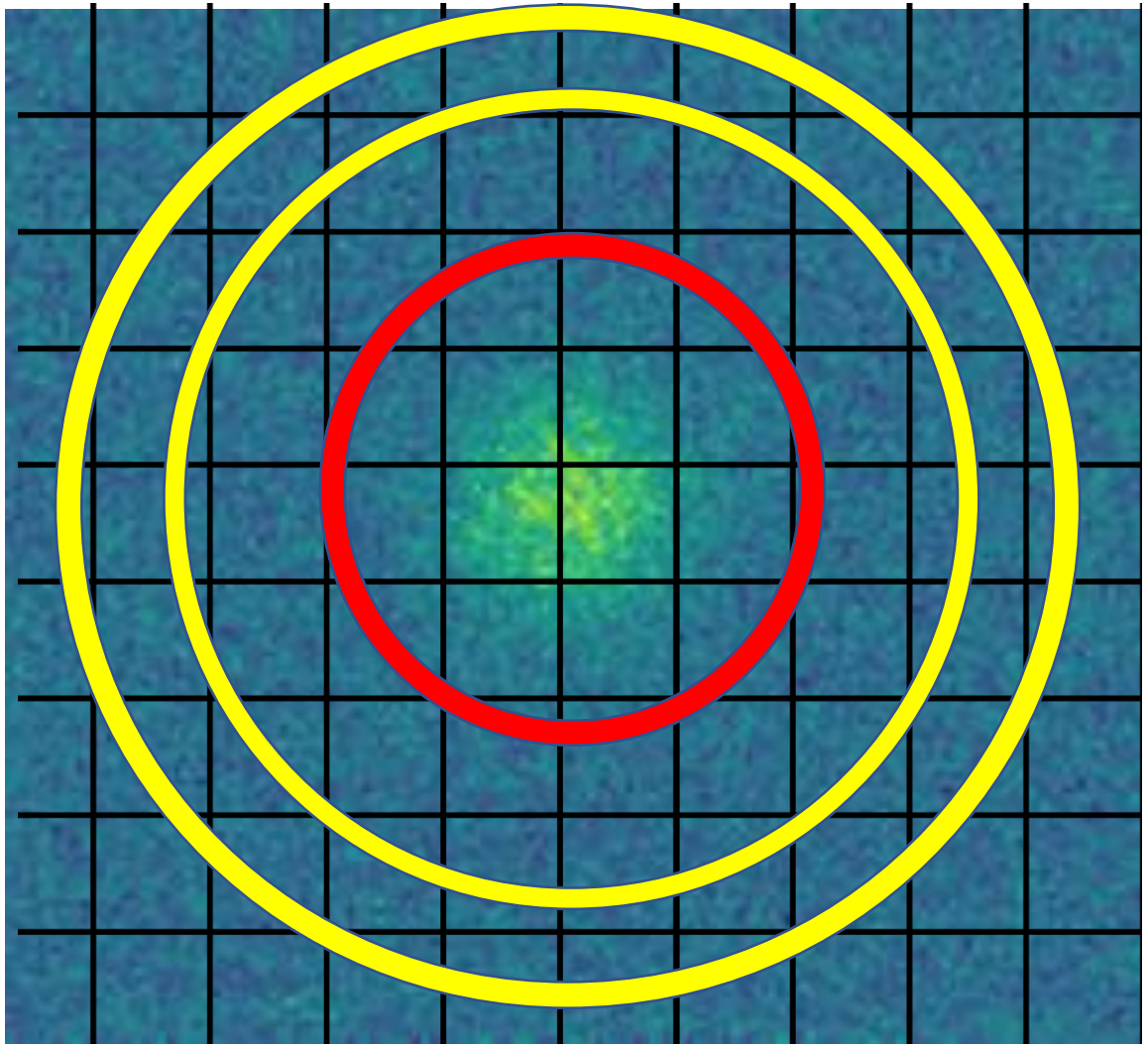
Which error term matters the most in this scenario?

$$SNR = \frac{gS_{star}}{\sqrt{gS_{star} + ngS_{sky} + ngS_{dark} + nS_{read}^2}}$$

S_{star} : Signal from the star (ADU)
S_{sky} : Signal from the sky (ADU)
S_{dark}: Signal from the dark (ADU)
S_{read}: Read noise (e-)
g: gain (e-/ADU)
n : Number of pixels

Aperture Photometry: Estimating Noise

Reduced Image

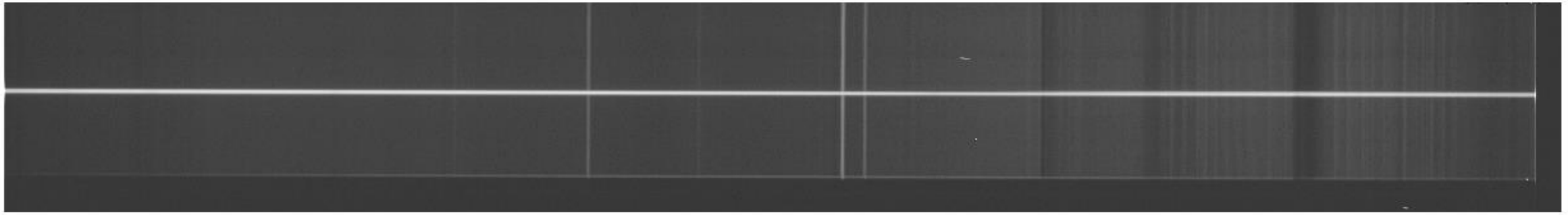


- You can choose an area away from your science target to estimate the residual random noise from the sky, dark current, etc.
- $$\mathrm{SNR} \sim \frac{\mathrm{Counts}_{\mathrm{star}}}{\sqrt{\mathrm{Counts}_{\mathrm{star}} + \mathrm{n}\sigma^2{}_{\mathrm{annulus}}}}$$
- You can check if your empirical errors match the previous SNR equation.

Spectroscopy: Estimating Noise

<- Spectral ->

Spatial ->



<- Spectral ->

Spatial ->



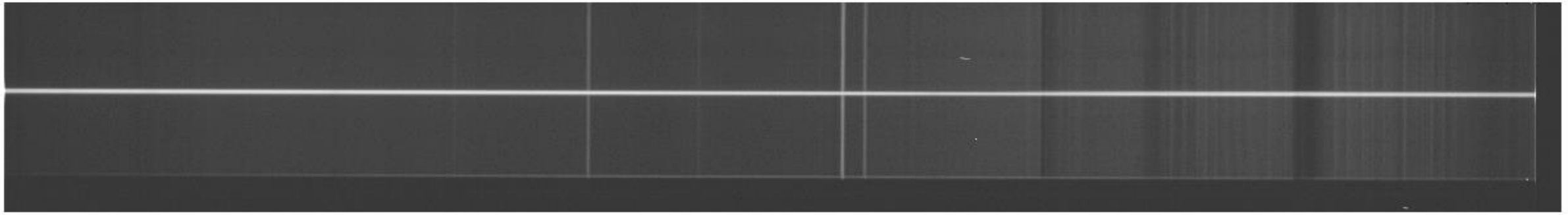
You add up light in the spatial direction to “extract” spectra (red).

You can take the standard deviation in the “background” (yellow) to estimate noise per wavelength.

Spectroscopy: Estimating Noise

<- Spectral ->

Spatial ->



<- Spectral ->

Spatial ->



$$\mathrm{SNR}\ (\lambda) \sim \frac{\mathrm{Counts}(\lambda)_{\mathrm{star}}}{\sqrt{\mathrm{Counts}(\lambda)_{\mathrm{star}} + \mathrm{n}\sigma(\lambda)^2_{\mathrm{annulus}}}}$$